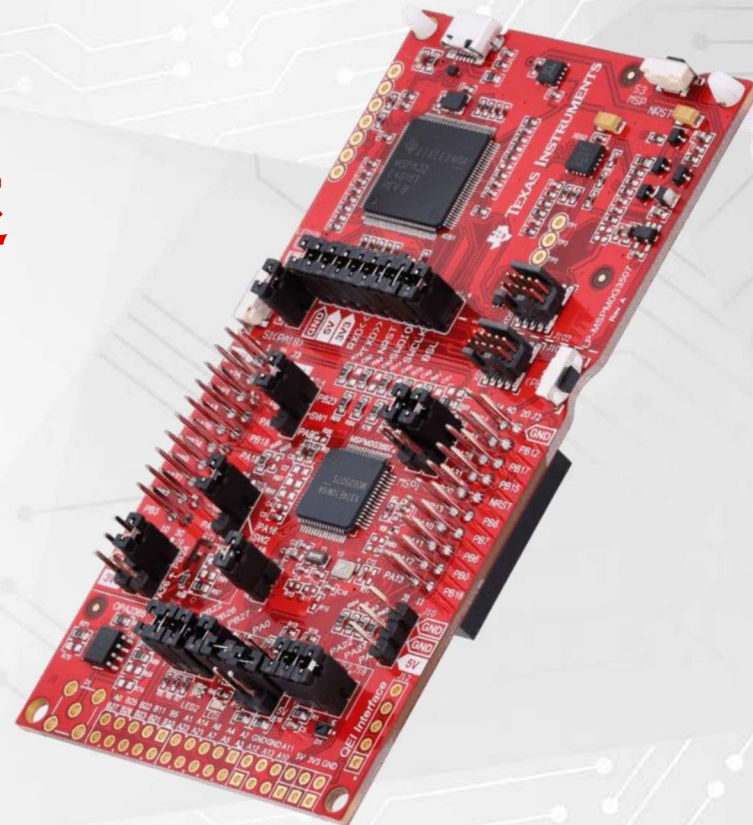


本文档根据TI官网手册整理，如有错误，[请参考文档slau869](#)

LP-MSPM0G3507硬件描述

- 谢胜祥
- 31342592@qq.com
- 2024.1.29



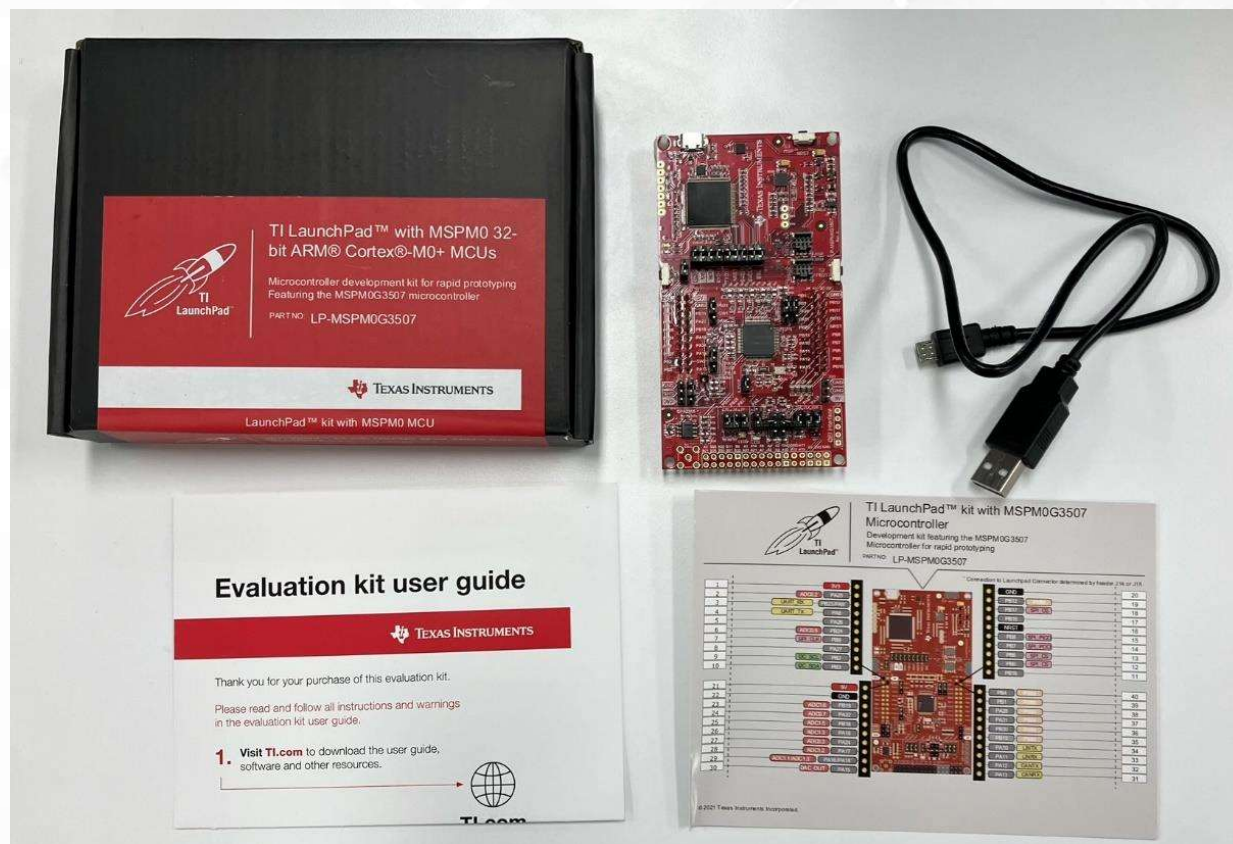
内容概述

- Launchpad套件概述
 - 板卡特性/MCU特性
- 调试器
- 系统供电
- 具体功能电路
 - MCU小系统
 - 按键、LED、
- 接口和跳线帽小结

Launchpad套件

LP-MSPM0G3507套件包含:

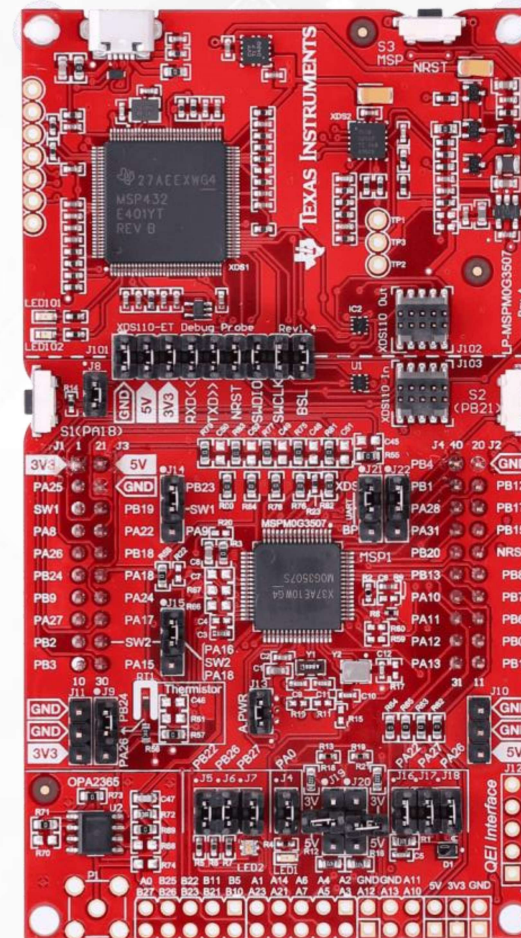
- MSPM0G3507 LaunchPad 板卡 x1
- Micro USB 数据线 x1
- 《快速入门指南》纸卡 x1



LP-MSPM0G3507特性

基本特性

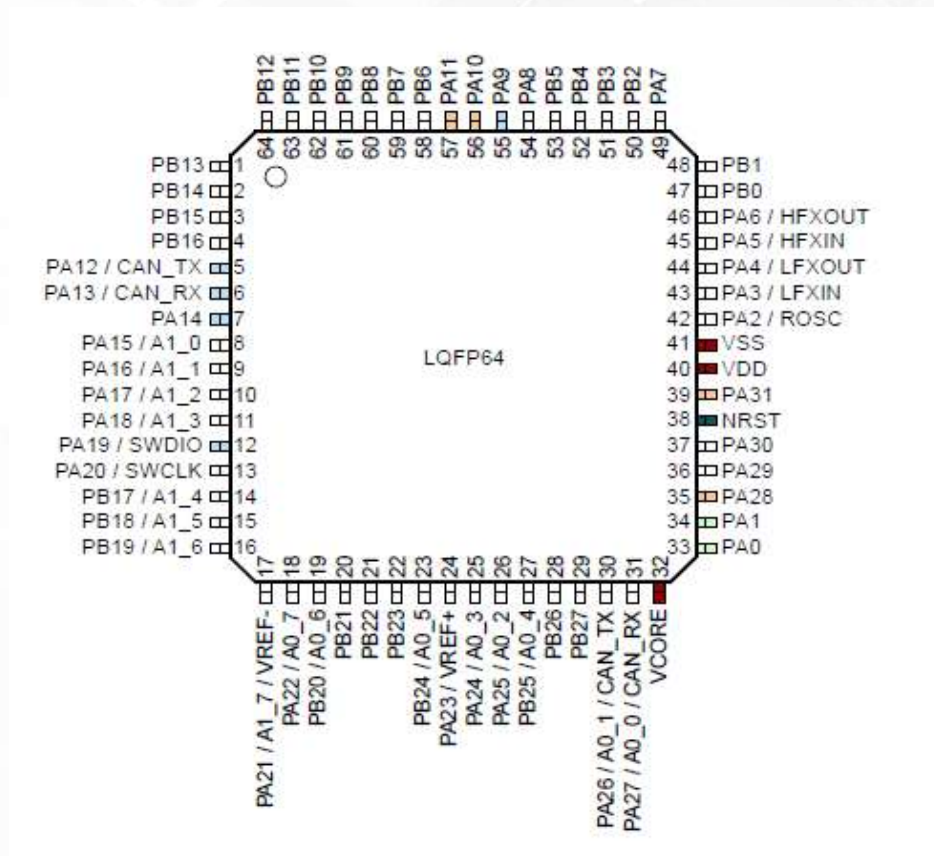
- 带外扩接口的板载 XDS110 -ET调试器
 - 可用于超低功耗调试的EnergyTrace 技术（配合CCS使用）
 - USB到PC的虚拟UART（USB-TTL转换器）
- 2 个用户按键(S1和S2)
- 1 个单色 LED1 和 1 个 RGB三色LED2
- 温度传感器电路（热敏电阻）
- 光传感器电路
- OPA2365电路（默认缓冲模式）可用于评估 4MSPS 的ADC
- ADC 输入的 RC 滤波器（默认未焊接）
- 所有引脚都引出



MCU—MSPM0G3507

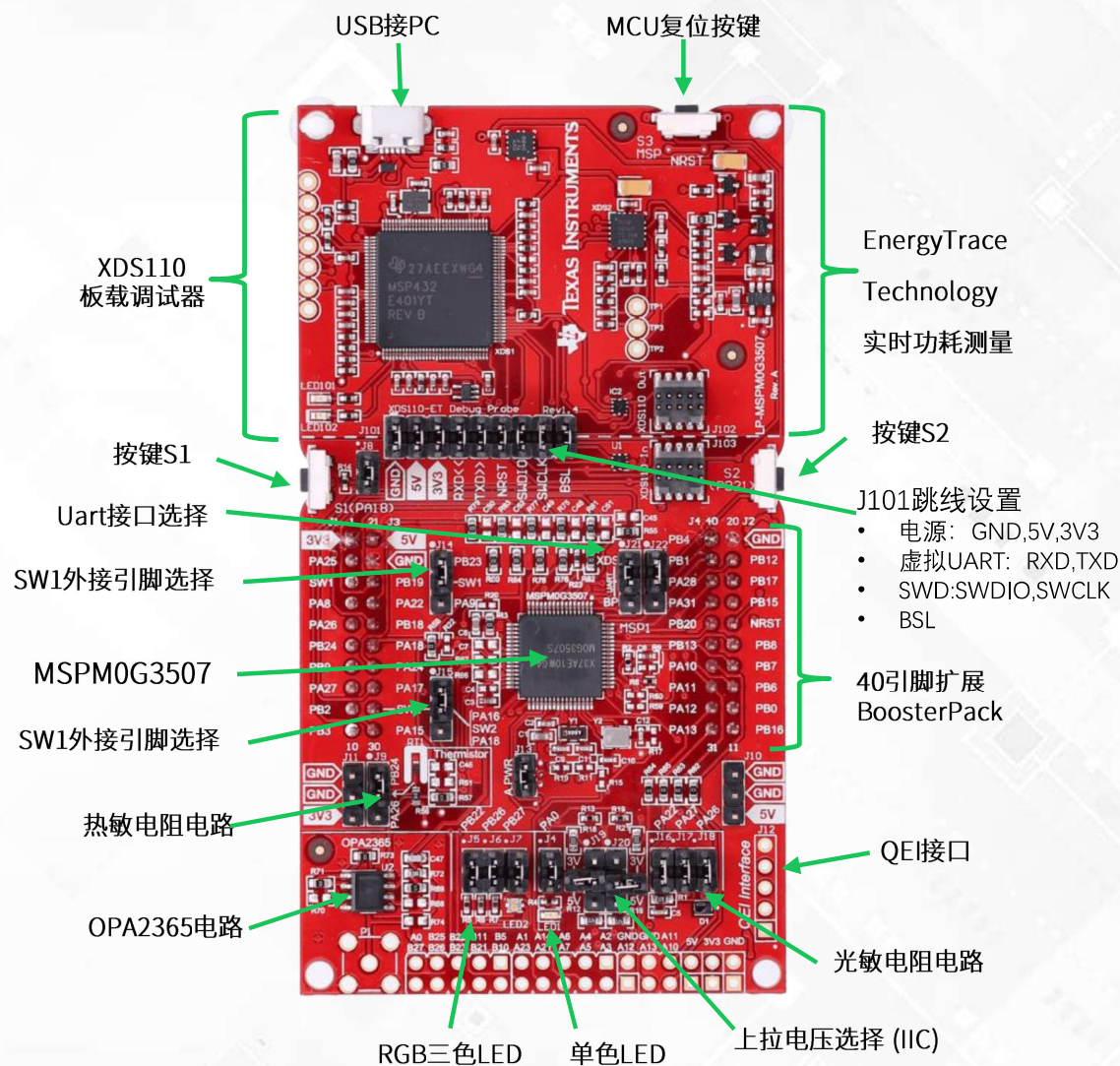
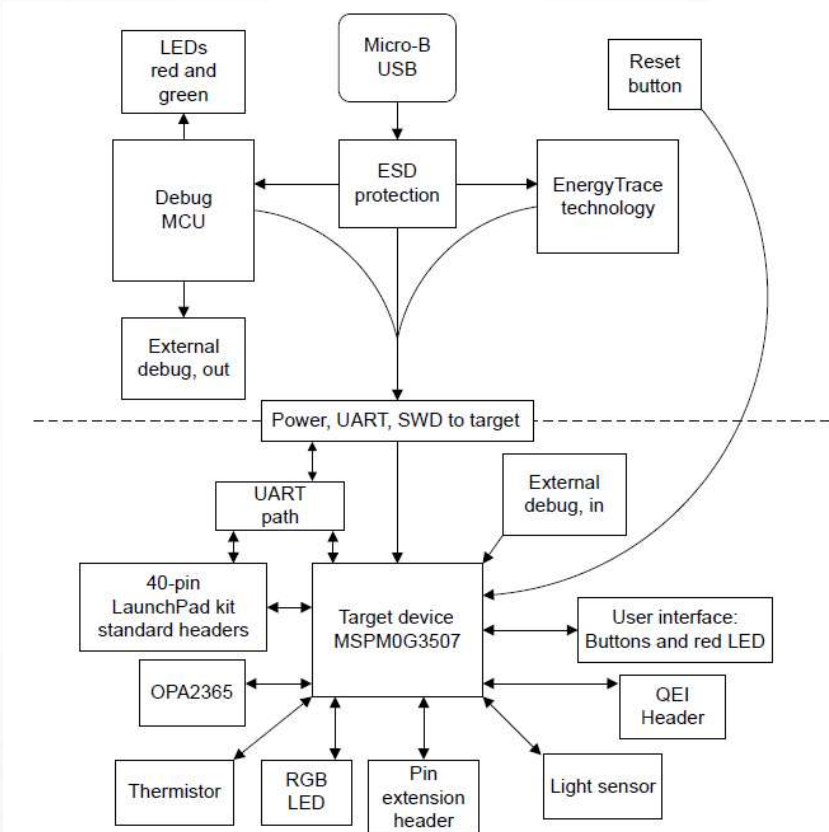
芯片特性：

- Arm 32 位 Cortex-M0+内核，最高80MHz
- 宽电源电压范围：1.62 V 至 3.6 V
- 工作温度范围：-40°C ~ 125 °C
- 具有纠错码128KB 闪存，32KB SRAM
- 2个总计多达17个外部通道的12 位 4Msps同步采样ADC
- 1个具有集成输出缓冲器的 12 位 1Msps DAC
- 2个零漂移、零交叉斩波运算放大器OPA
- 1个通用运算放大器GPAMP
- 3个带8位DAC参考的高速比较器
- 内部参考电压（1.4V或2.5V）
- 4个UART，2个IIC，2个SPI，1个CAN
- 7个 定时器，2个窗口式看门狗定时器，RTC
- 60个 GPIO

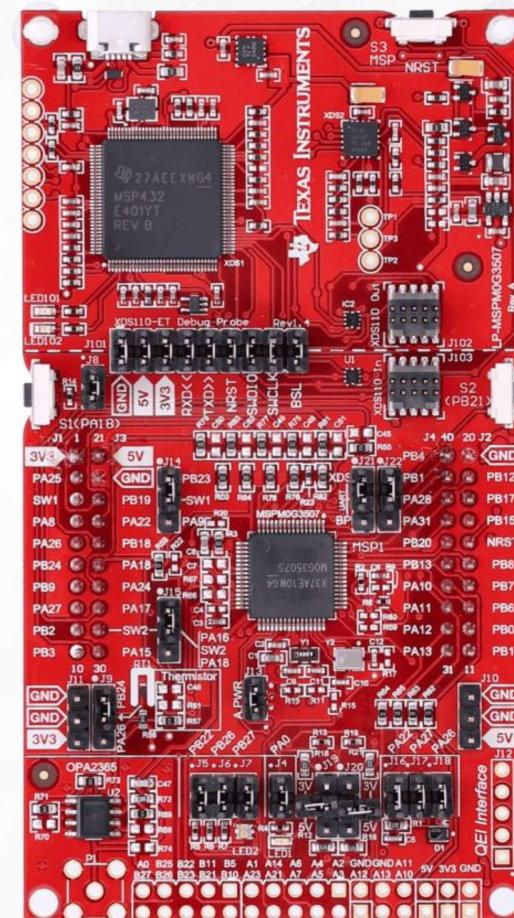
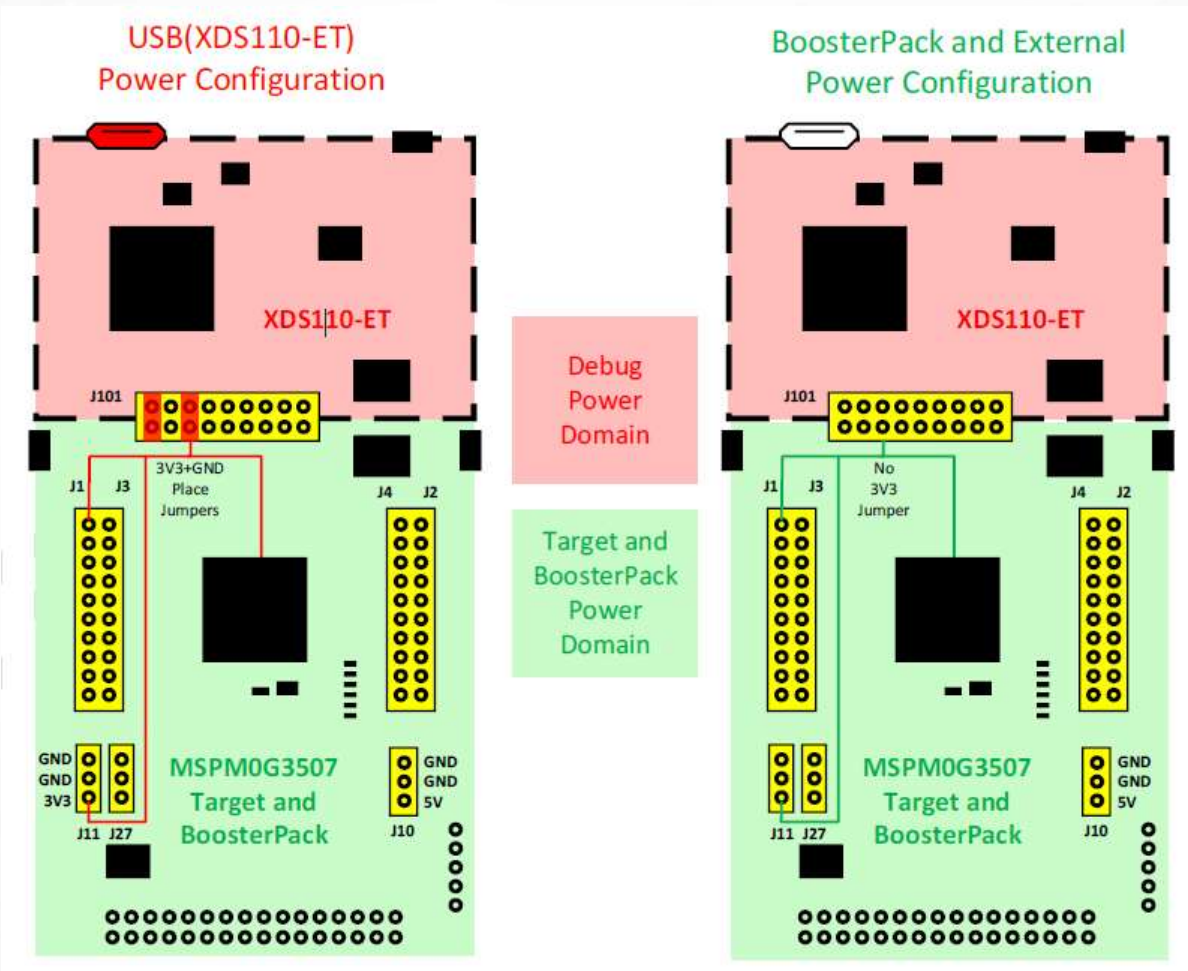


更多特性描述请查考器件手册

功能框图



板卡供电描述--电源(3.3V)

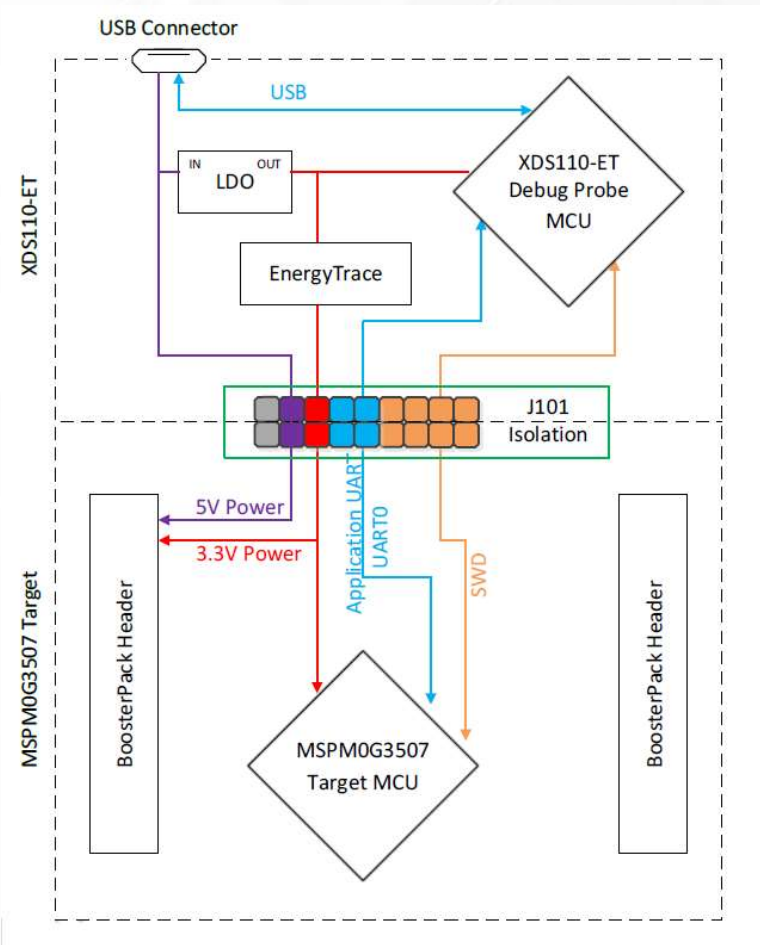


板载调试器

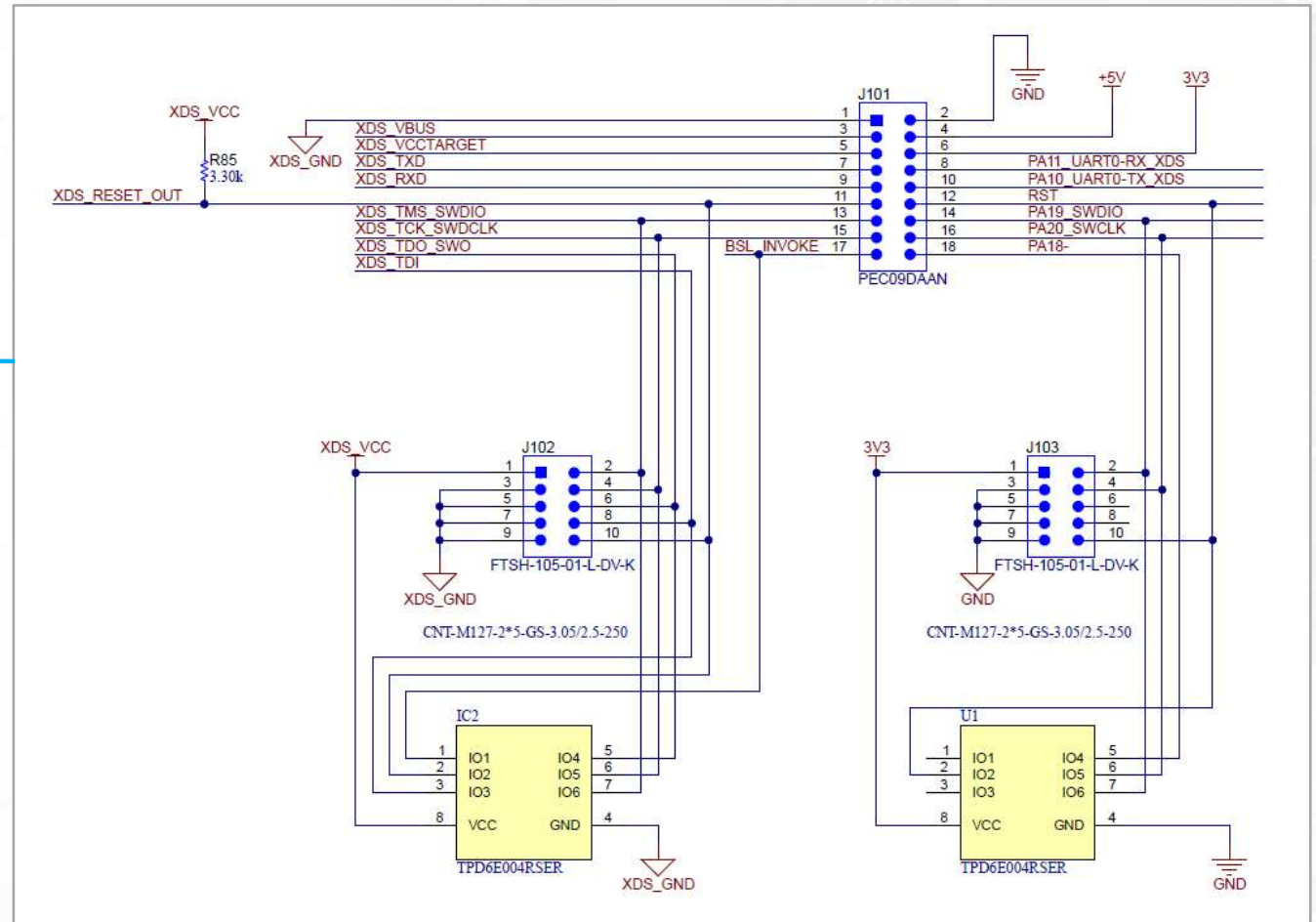
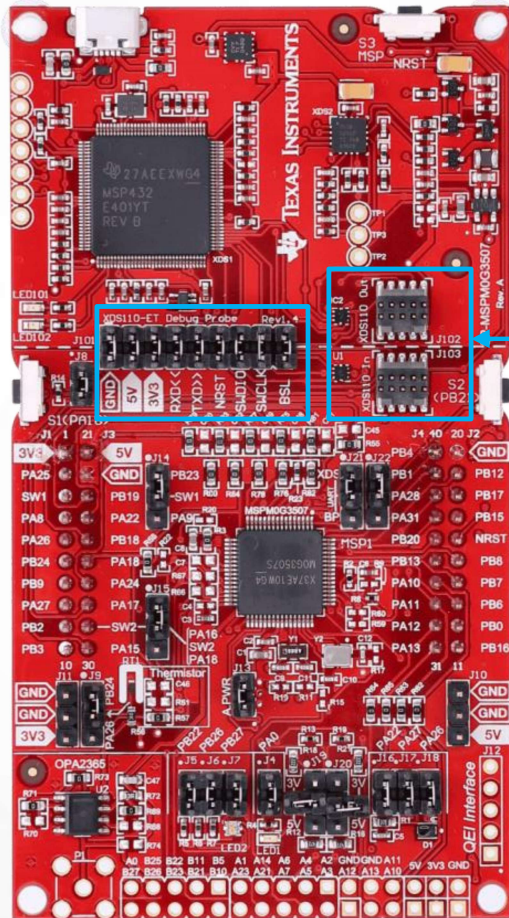
- 支持EnergyTrace技术的 XDS110-ET 板载调试器
 - 调试器连接：跳线帽的设置
- XDS110-ET调试器对外使用（SWD）
- XDS110-ET调试器用于外部板卡(JTAG)
- 使用外部调试器代替板载 XDS110-ET

跳线	描述
GND	地
5V	来自USB的VBUS 5V
3V3	3.3V，从XDS110-ET域的VBUS出取得
RXD<<	虚拟UART: MSPM0G3507从该引脚接收数据
TXD>>	虚拟UART: MSPM0G3507发送数据到该引脚
NRST	RST信号
SWDIO	SWD调试信号：SWDIO数据信号
SWCLK	SWD调试信号：SWCLK时钟信号
BSL	Bootstrap Loader信号

J101接口描述

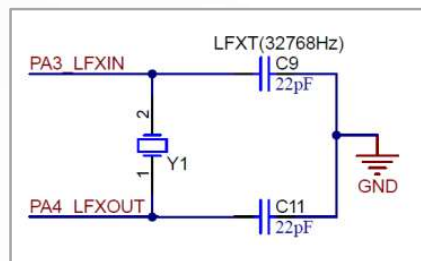


调试器接口

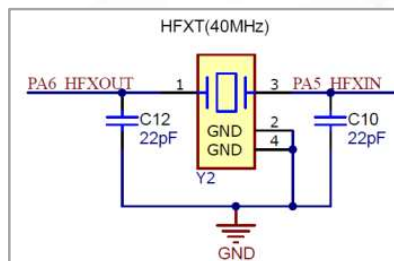


MCU小系统

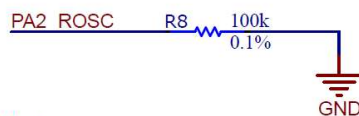
低频晶振



高频晶振



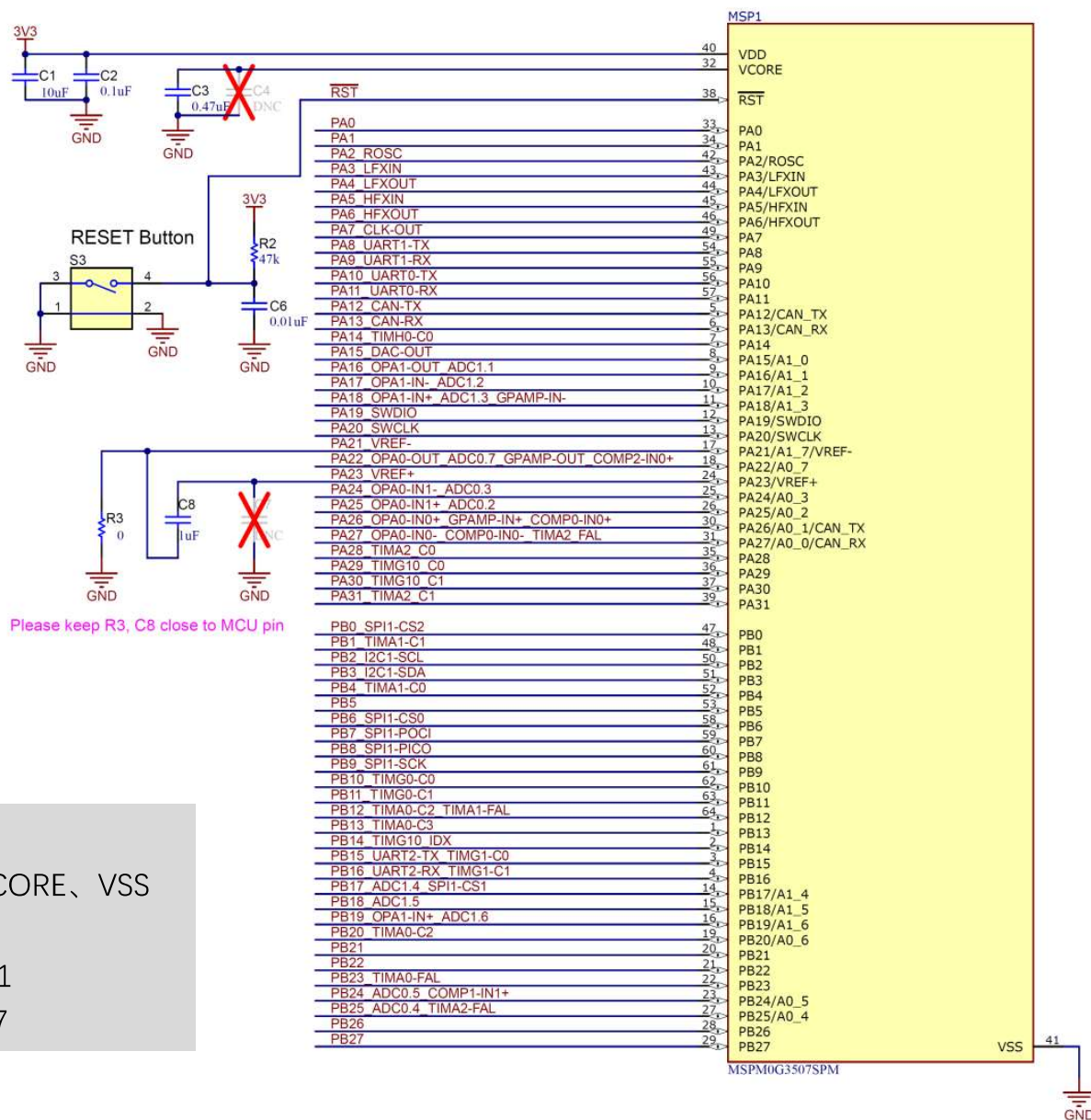
ROSC



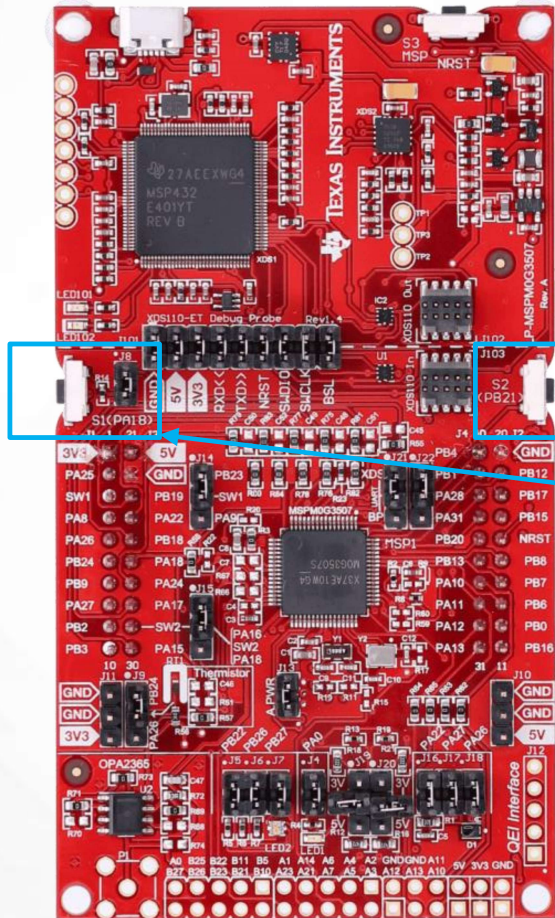
Please keep R8 close to MCU pin

芯片共64个引脚:

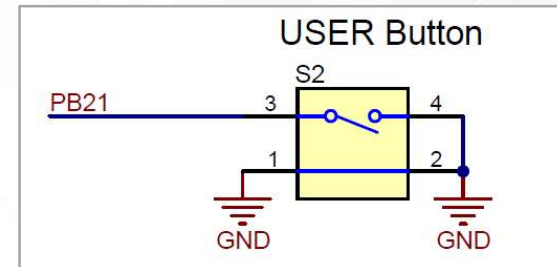
- 3个: VDD、VCORE、VSS
- 1个: RST
- 32个: PA0-PA31
- 28个: PB0-PB27



按键

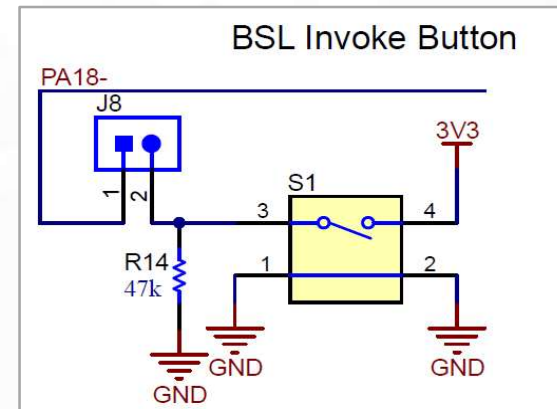


用户按键S2



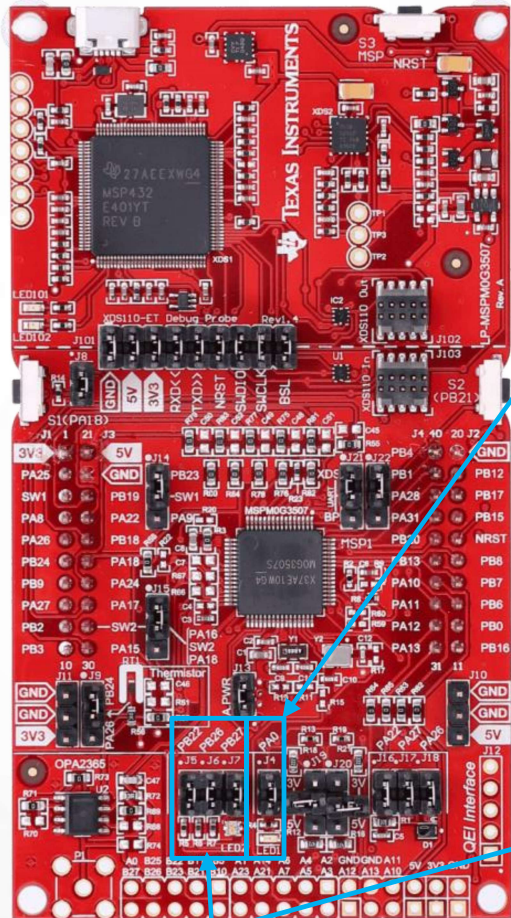
PB21配置输入，还需配置内部上拉，按键未按下时为高电平；按键按下时为低电平

BSL按键S1

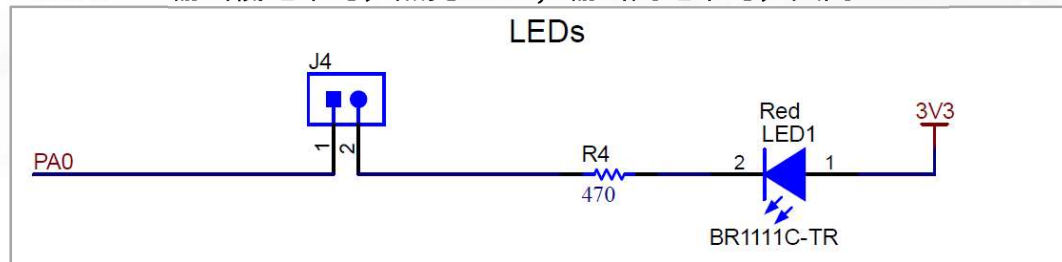


不用使能内部电阻，S1按键未按下时为低电平；按键按下时为高电平

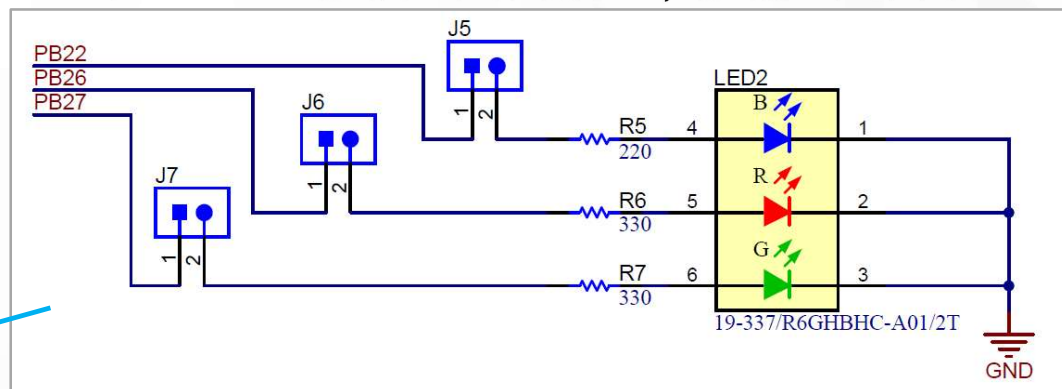
LEDs



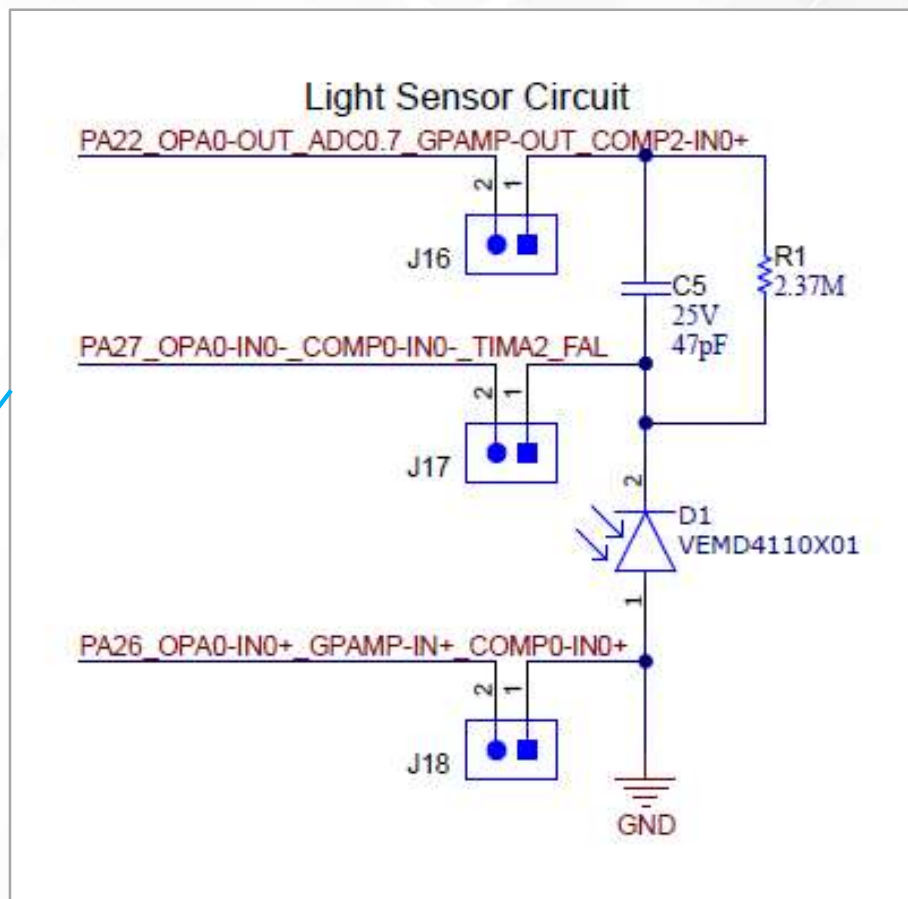
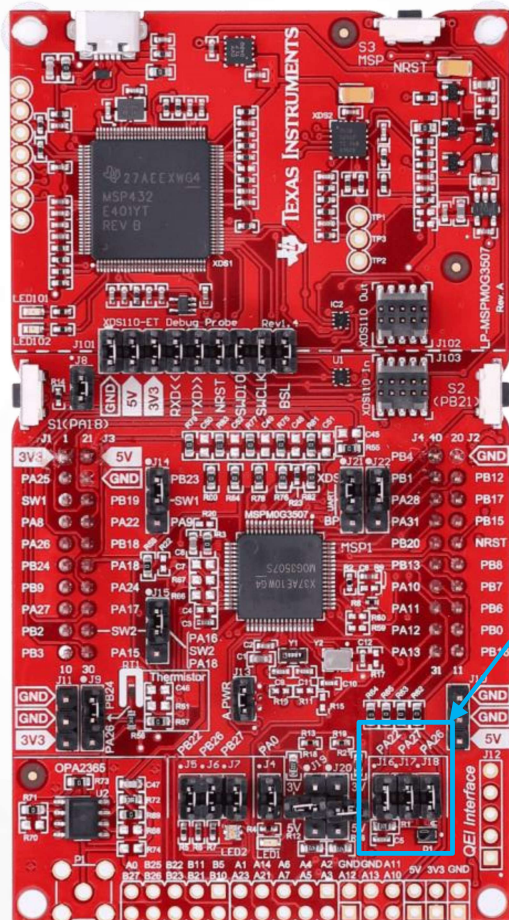
LED1 输出低电平时，点亮LED1；输出高电平时，关闭LED1



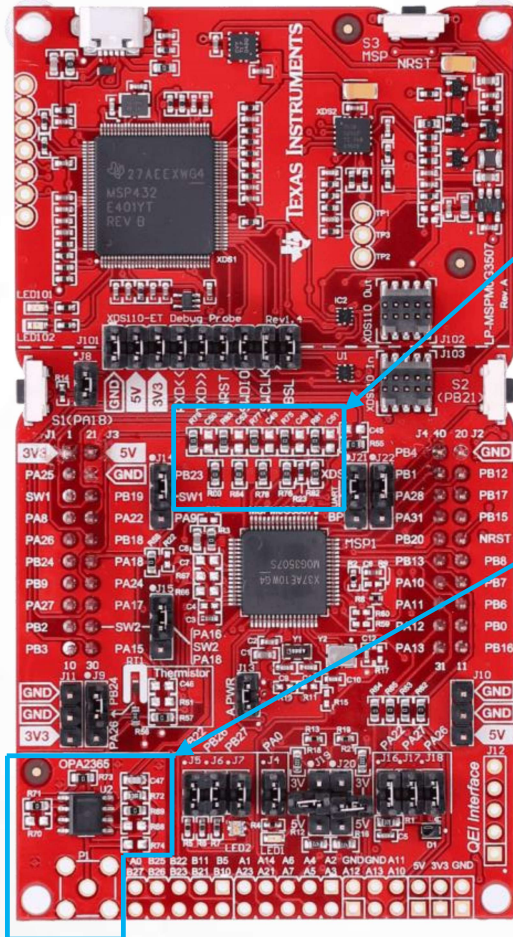
LED2,三色LED 输出高电平时，点亮LED；输出低电平时，关闭LED



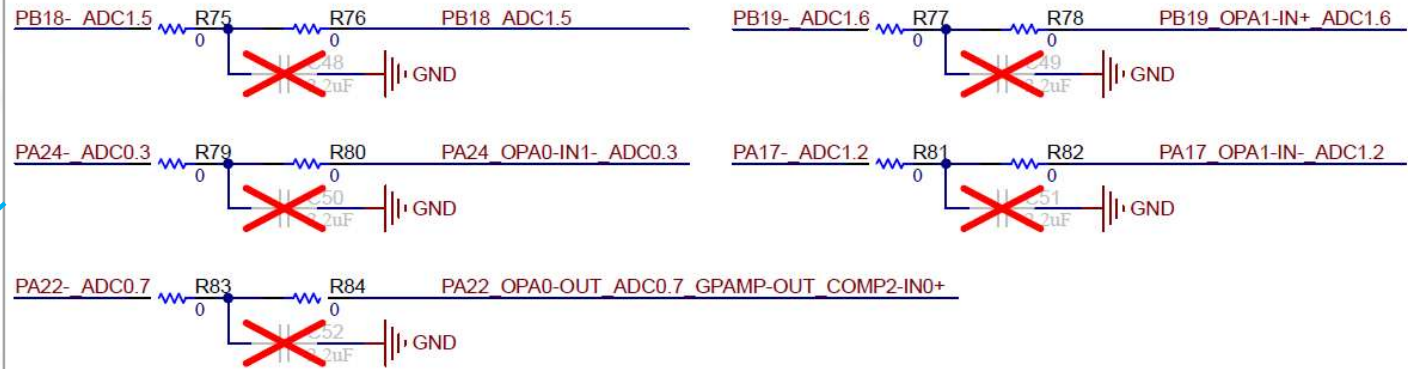
光照度传感器电路



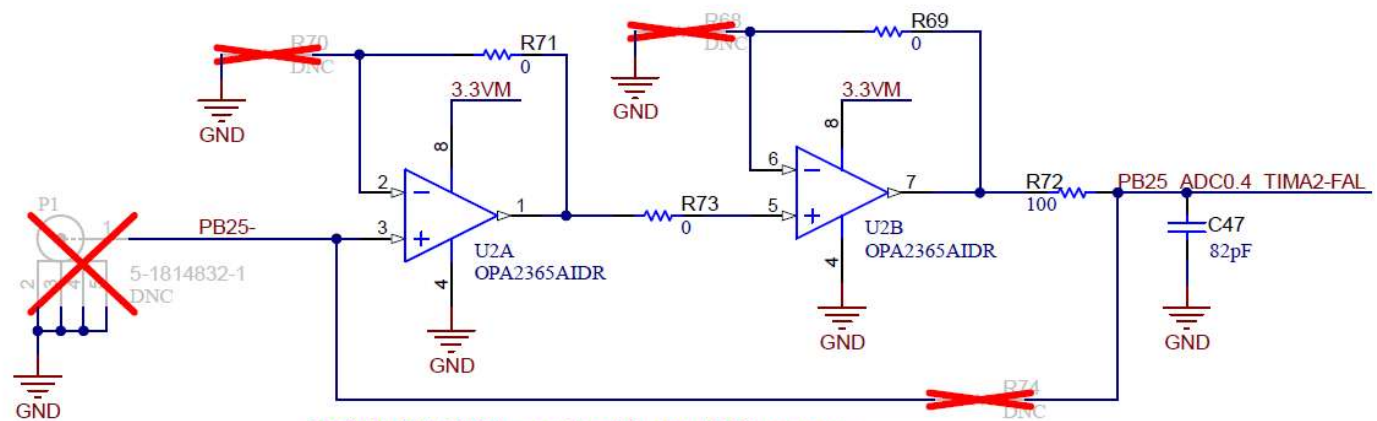
ADC测试电路



RC Filter for ADC Input

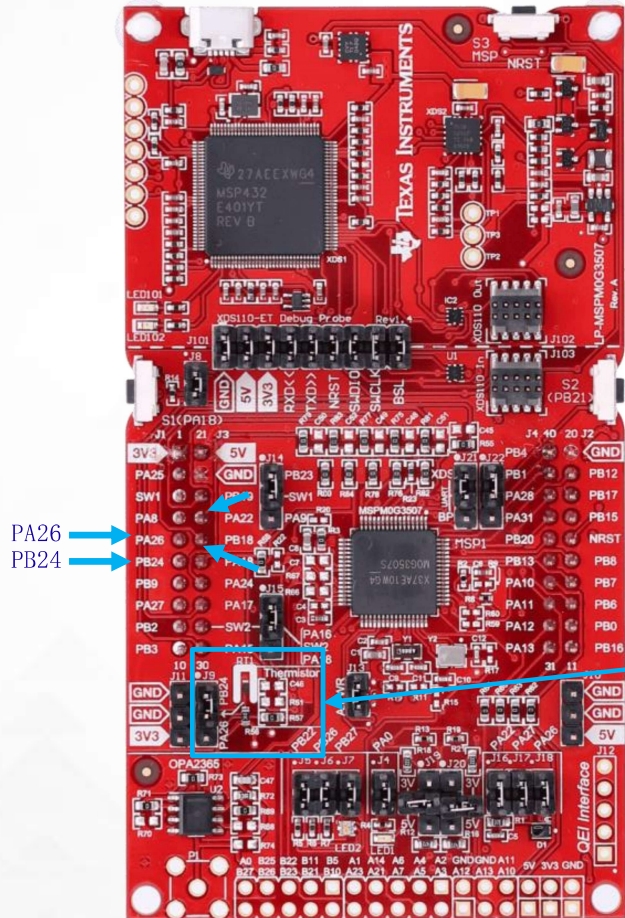


ADC Input with Active Buffer



Default OPA2365 is populated for the ADC input test;
If OPA2365 is NOT used, R72 need to be taken off and R74 0-ohm to be populated;

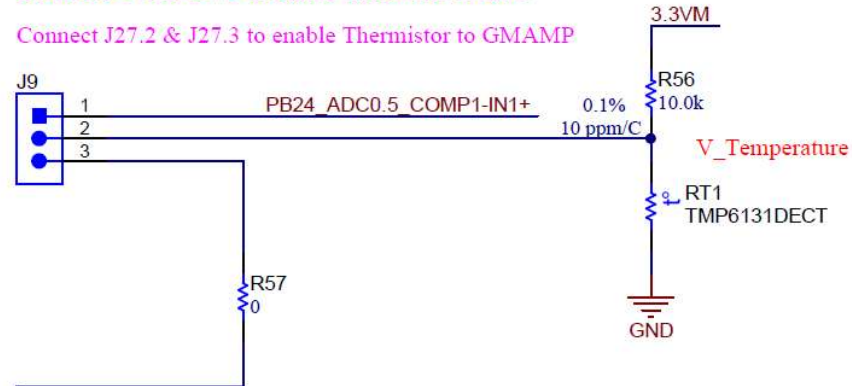
热敏电阻和通用运放



Thermistor Circuit

Connect J27.1 & J27.2 to enable Thermistor to ADC

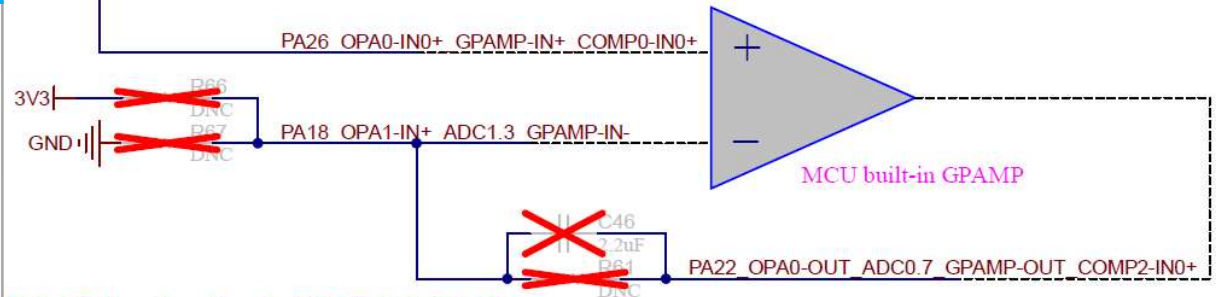
Connect J27.2 & J27.3 to enable Thermistor to GMAMP



GPAMP can work in Buffer mode or Amplify mode via change R57, R61, R66, R67, C46

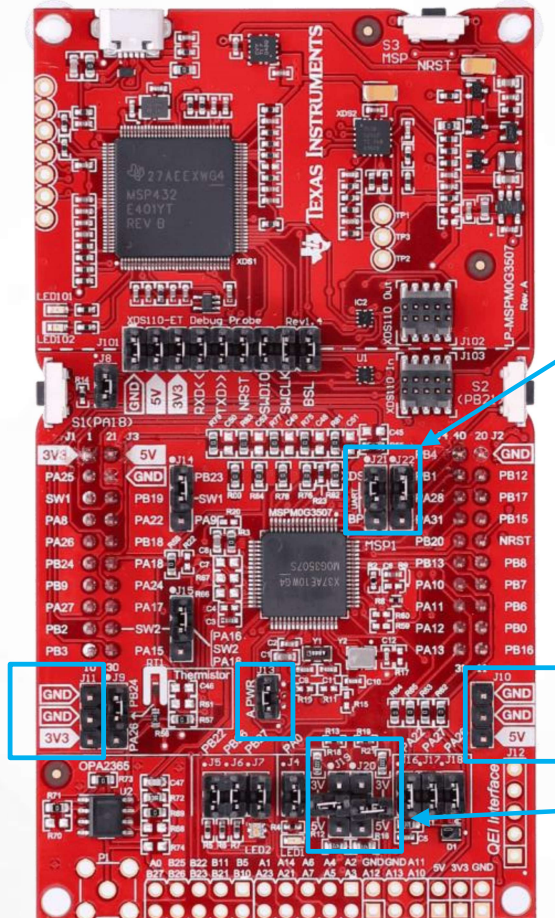
GPAMP Test Circuit

This GPAMP can used in Thermistor or Motor Control(3 ISEN algrithom)



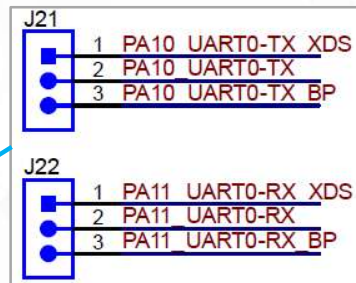
R66, R67 is voltage bias for GPAMP, 100-300mV

跳线帽设置



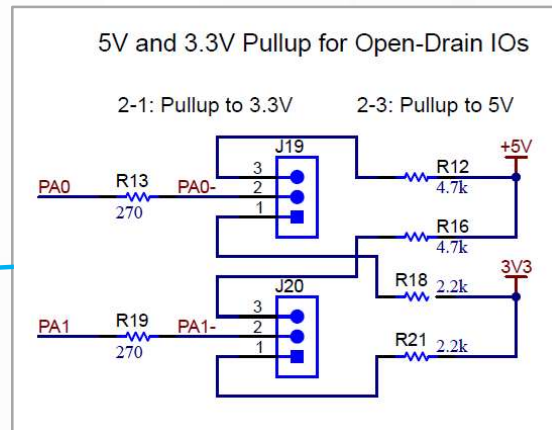
芯片的uart引脚通过跳线:

- 连接到xds
- 连接到可以BP

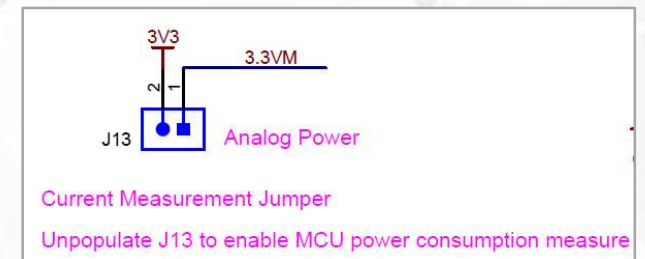
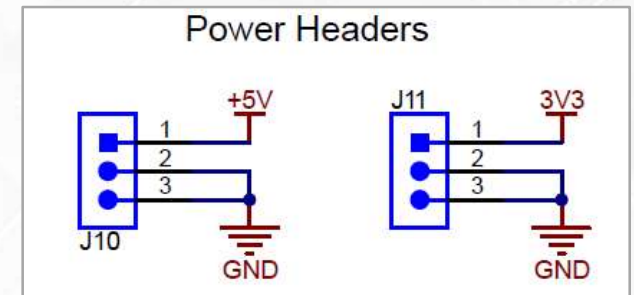


开漏输出的引脚选择上拉电源

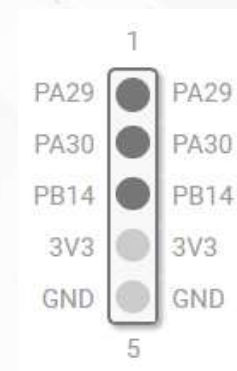
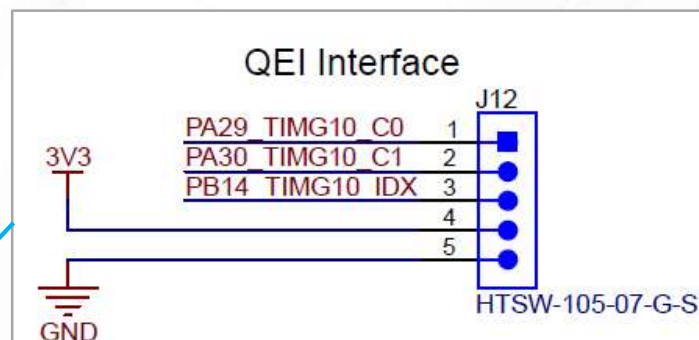
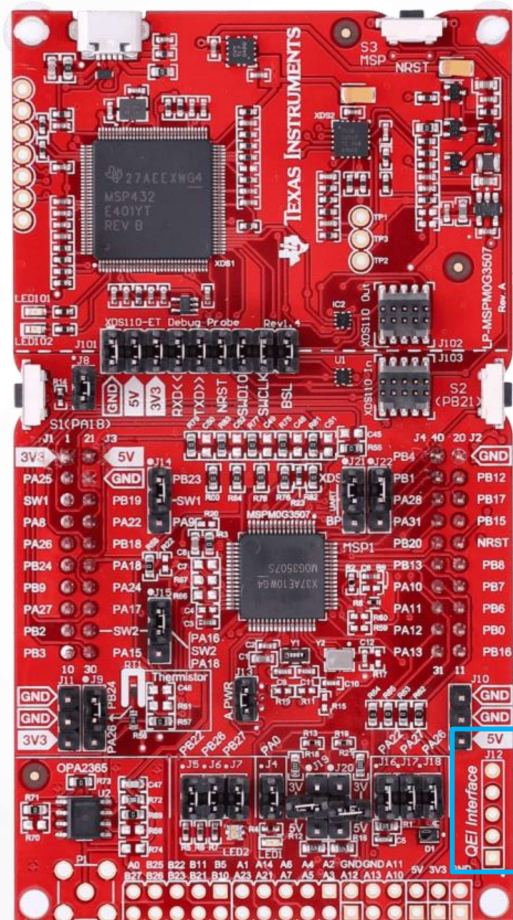
- 3.3V
- 5V



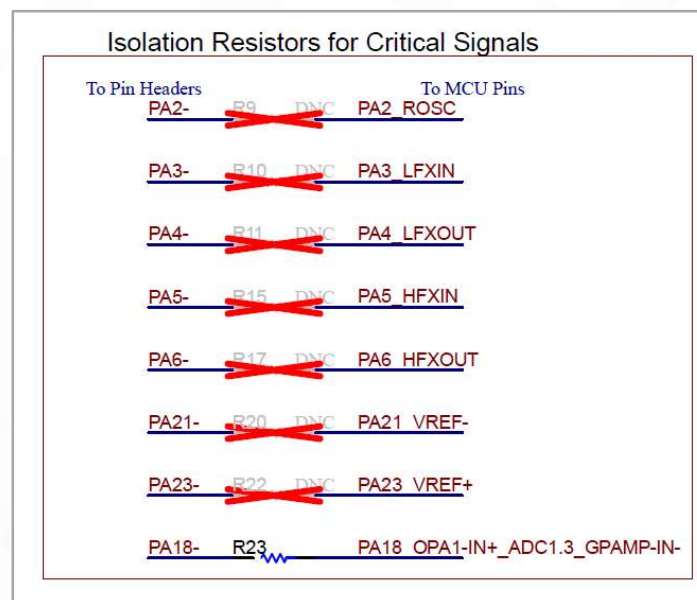
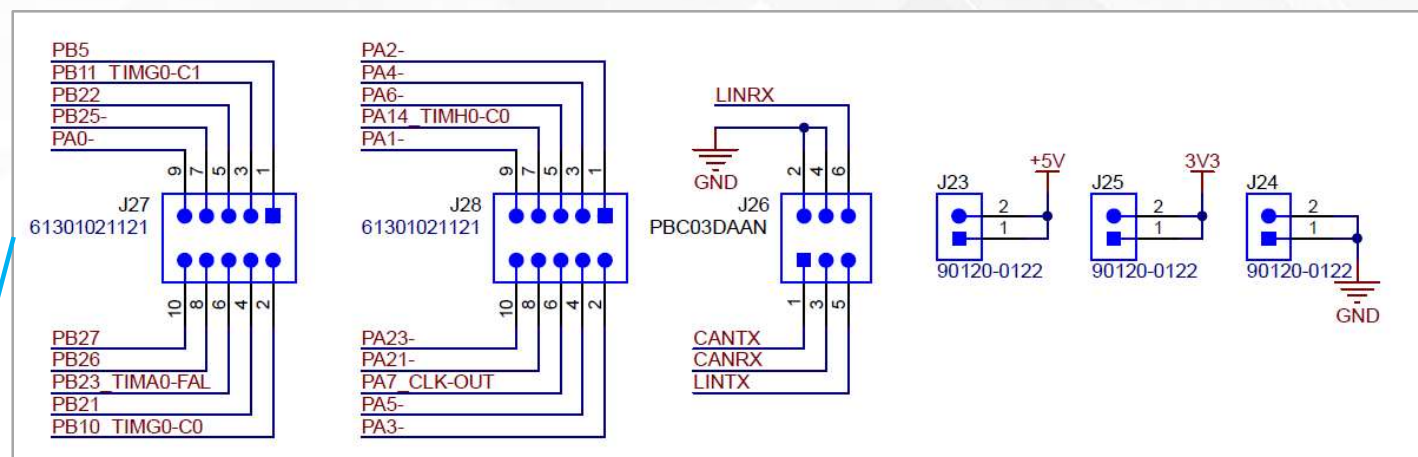
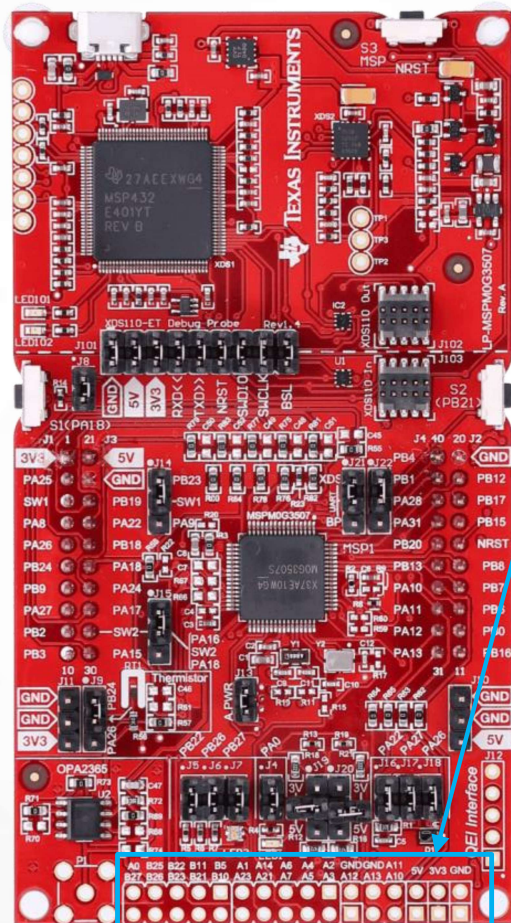
电源接口



QEI接口



底部的扩展接口



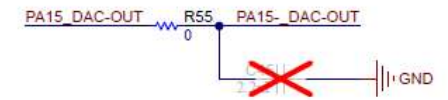
部分特殊功能引脚也对外引出到底部的接口，通过0欧电阻配置。

默认0欧电阻未焊接，是断开的。

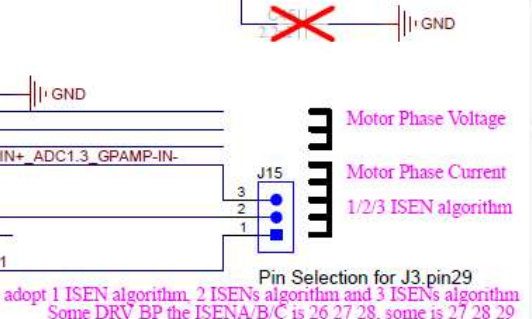
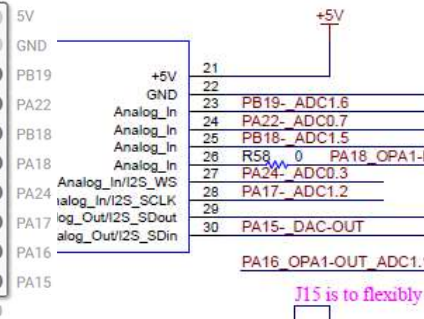
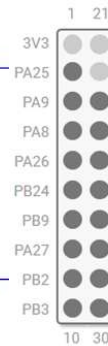
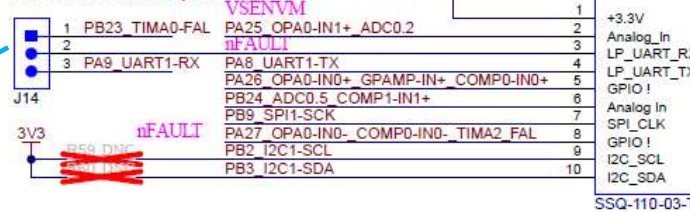
BoosterPack接口

BoosterPack Connectors

RC Filter for DAC Output



Pin Selection for J1.pin3

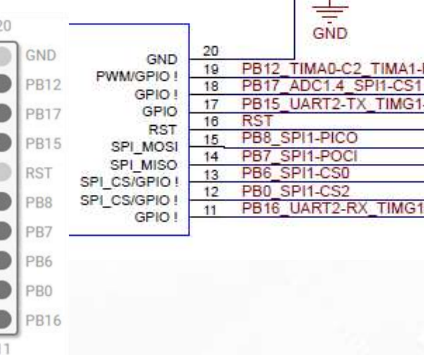
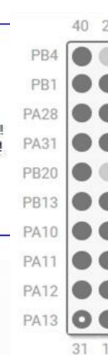
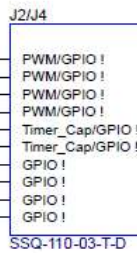
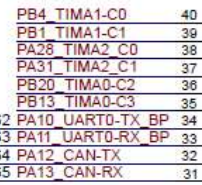


Pin Selection for J3.pin29

J15 is to flexibly adopt 1 ISEN algorithm, 2 ISENs algorithm and 3 ISENs algorithm
Some DRV BP the ISENA/B/C is 26 27 28, some is 27 28 29

6*PWM

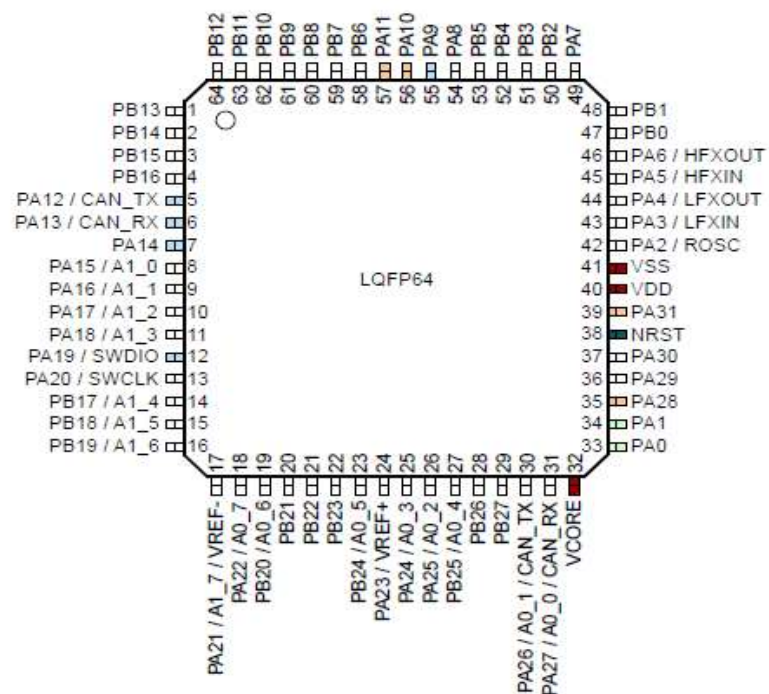
CAN/LIN



HALL-A or SPI
HALL-B or SPI

SPI
HALL-C or SPI
SPI

接口描述



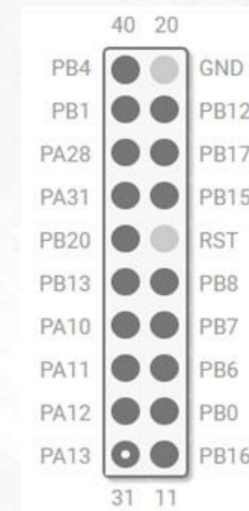
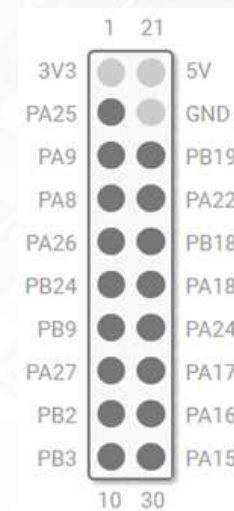
芯片共64个引脚:

3个: VDD、VCORE、VSS

1个: RST

32个: PA0-PA31

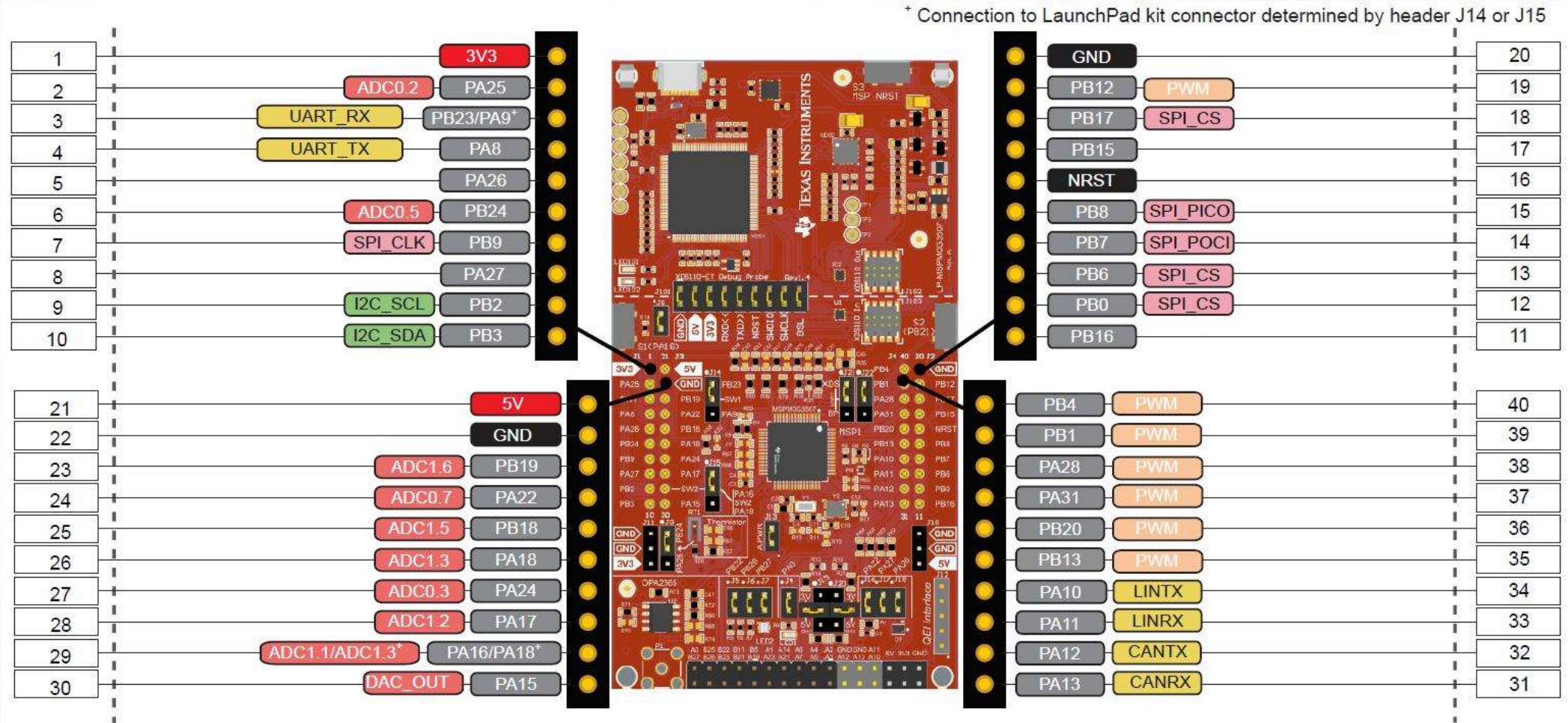
28个: PB0-PB27



所有的IO引脚都引出来了

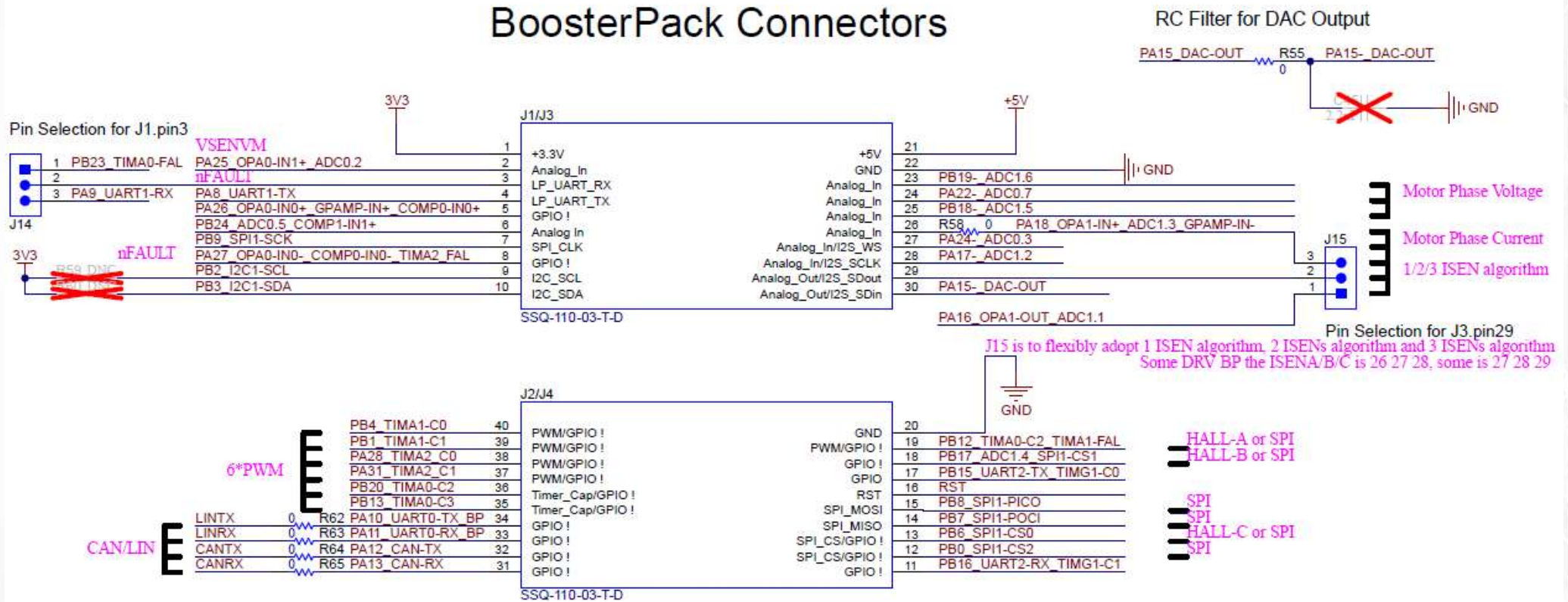
但是功能有复用, 需参考具体电路

BoosterPack接口



BoosterPack接口

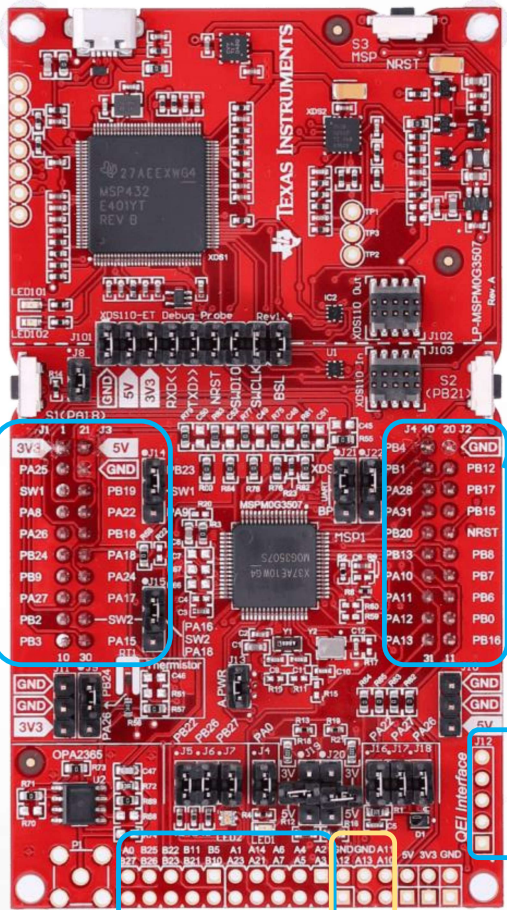
BoosterPack Connectors



引脚示意图

BP接口

3V3	1	21	5V
PA25			GND
PA9			PB19
PA8			PA22
PA26			PB18
PB24			PA18
PB9			PA24
PA27			PA17
PB2			PA16
PB3			PA15
	10	30	



BP接口

	40	20	
PB4			GND
PB1			PB12
PA28			PB17
PA31			PB15
PB20			RST
PB13			PB8
PA10			PB7
PA11			PB6
PA12			PB0
PA13			PB16
	31	11	

1	PA29
	PA30
	PB14
	3V3
	GND
5	

QE接口

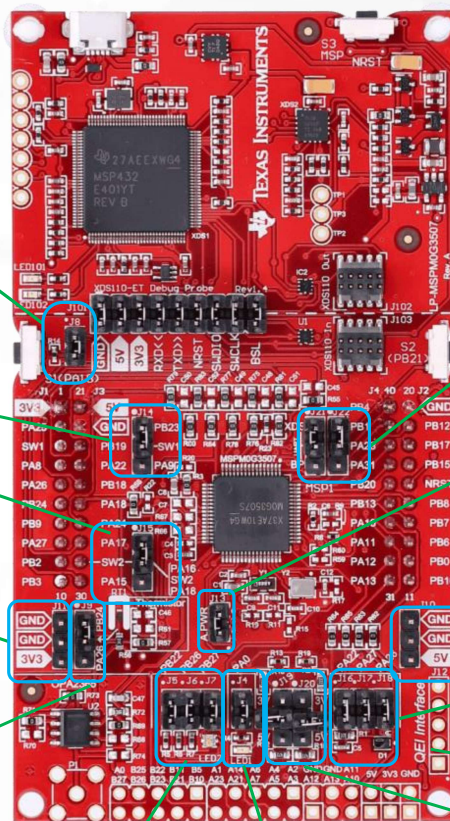
扩展接口

CAN/LIN接口

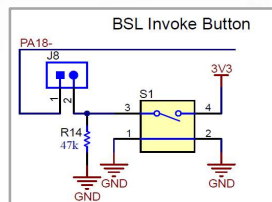
PA0 → J4 → LED1; → J19 IIC上拉;
 PA1 → J20 IIC上拉
 PA2 → ROSC
 PA3 低频晶振
 PA4 低频晶振
 PA5 高频晶振
 PA6 高频晶振
 PA7
 PA8 → J16 TARGET_TXD
 PA9 → J17 TRARGET_RXD
 PA10 → J21, UART0_TX_XDS → LIN
 PA11 → J22, UART0_RX_XDS → LIN
 PA12 CAN
 PA13 CAN
 PA14
 PA15
 PA16 → J15
 PA17
 PA18 → J8 S1按键; → J15 → PA16
 PA19 SWDIO
 PA20 SWCLK
 PA21 VREF- → GND
 PA22 ← J16
 PA23 VREF+
 PA24
 PA25
 PA26 → J18 → GND RGB LED红色
 PA27 → J17 → 光传感器负端
 PA28
 PA29 → QE接口
 PA30 → QE接口
 PA31

PB0
 PB1
 PB2
 PB3
 PB4
 PB5
 PB6
 PB7
 PB8
 PB9
 PB10
 PB11
 PB12
 PB13
 PB14 → QE接口
 PB15
 PB16
 PB17
 PB18
 PB19
 PB20
 PB21 ← S2按键
 PB22 → J5 → RGB LED
 PB23 → J14 → BP
 PB24 ← J9 ← 热敏电阻
 PB25 ← OPA2365 ← 底部接口
 PB26 → J6 → RGB LED
 PB27 → J7 → RGB LED

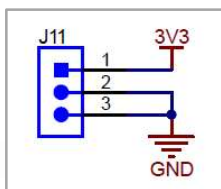
跳线配置说明



SW1和SW2引脚选择



J8



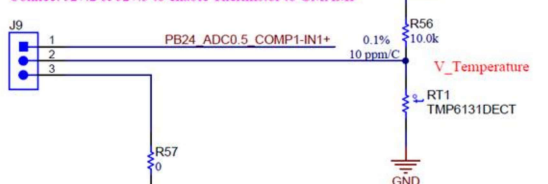
J14

J15

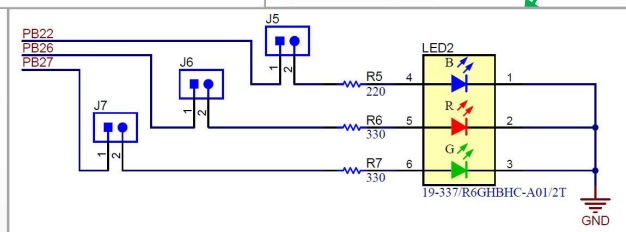
J11

Thermistor Circuit

Connect J27.1 & J27.2 to enable Thermistor to ADC
Connect J27.2 & J27.3 to enable Thermistor to GMAMP



J9



J5 J6 J7

J4

J21 J22

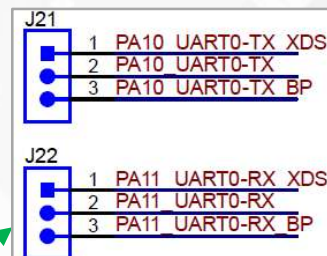
J13

J10

J16 J17 J18

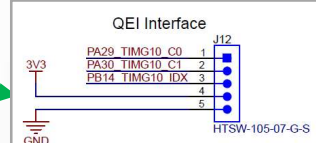
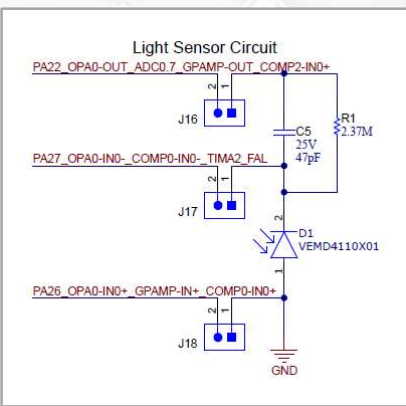
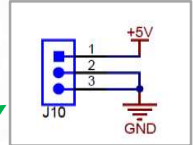
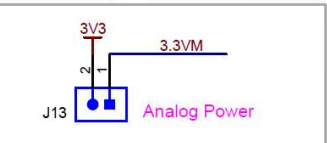
J12

J19 J20



芯片的uart引脚选择:

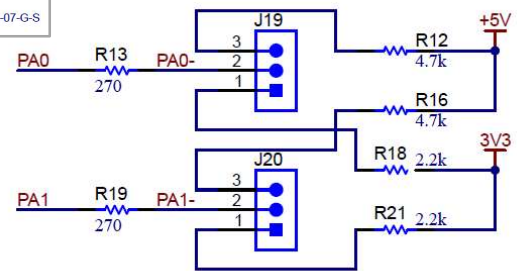
- 连到XDS110
- 连到BP排针



5V and 3.3V Pullup for Open-Drain IOs

2-1: Pullup to 3.3V

2-3: Pullup to 5V



谢谢!